

From the $C_1 \otimes B_2$ row to §4.2 of Lublinsky–Mulian

The p^+ integration done term by term — UV, finite and rapidity terms, all of them

arXiv:1610.03453v2, §4.2 and Appendices C, H.2 — one continuous derivation

0. The short version

Your row is the vertex of Eq. (4.16) of the paper. Under the identification

$$z_{\text{paper}} \leftrightarrow x, \quad z'_{\text{paper}} \leftrightarrow z, \quad y \leftrightarrow y$$

one has $Y = y - z_{\text{paper}} = y - x = s$, $Y' = y - z'_{\text{paper}} = y - z = s - r$, $Z = z_{\text{paper}} - z'_{\text{paper}} = x - z = -r$, and therefore

$$\bar{\xi} D = \bar{\xi} (s - r)^2 + \xi s^2 = (1 - \xi)(Y')^2 + \xi Y^2,$$

which is *exactly* the second energy denominator of (4.16). Better still, the six ξ -weights of your bracket are exactly the six products that the paper's vertex Λ, Θ decomposition produces. So the whole of §4.2 / Appendix H.2 applies, and the p^+ integral is elementary: it needs only the paper's (C.2), (C.5), (C.6). Doing it gives

$$\int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi \frac{\text{bracket}}{\bar{\xi} D} = \underbrace{\ell_k \left[\frac{V_2}{Y^2} + \frac{V_4}{(Y')^2} \right]}_{\text{(a) rapidity} \rightarrow (\delta Y)^2, \text{ Eq. (H.7)}} + \underbrace{\ln \frac{Y^2}{(Y')^2} \mathcal{G}_{\log}}_{\text{(b) the NLO logs, Eq. (H.6)}} + \underbrace{\mathcal{G}_{\text{rat}}}_{\text{(c) rational, Eq. (H.6)}},$$

with $\ell_k \equiv \ln(k^+/\Lambda)$, and the ultraviolet ($z \rightarrow x$) content of the whole thing sitting in one place:

$$\left[\int d\xi \frac{\text{bracket}}{\bar{\xi} D} \right]_{r \rightarrow 0} = \left(2\ell_k - \frac{11}{6} \right) \frac{(s \cdot P)}{P^2 s^2} \frac{1}{r^2} + (\text{integrable}), \quad 2\ell_k - \frac{11}{6} = \int d\xi \frac{P_{gg}(\xi)}{2N_c}.$$

The $2\ell_k$ comes entirely from block (a) and the $-\frac{11}{6}$ entirely from blocks (b)+(c) — verified numerically in §10E. And $-\frac{11}{6} \times 2N_c = -\frac{11N_c}{3} = -b$ at $N_f = 0$: this row is a running-coupling contribution. §9 shows that its ξ integral is *literally* the paper's (C.10), which is where $\frac{67}{18} - \frac{\pi^2}{3}$ comes from.

1. Three things in the expression as typed

These have to be settled before any integral, because two of them change the answer.

(i) Θ_6 carries a minus sign twice. In (1.1) the last structure already appears with an overall minus, and your (1.10) gives the result of contracting it:

$$-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2} - \frac{(y-x)^m(y-z)^i}{2(y-z)^2} + \frac{(y-z)^m(x-z)^i}{(x-z)^2} + \frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right] \frac{P^i r^m}{P^2 r^2} = + \frac{2(P \cdot r)(r \cdot s)}{P^2 r^4} - \frac{P \cdot r}{P^2 r^2} + \dots$$

In the expression you typed this same right-hand side appears again behind a minus sign, $-\left[\frac{2(P \cdot r)(r \cdot s)}{P^2 r^4} - \dots\right]$. That flips Θ_6 . It matters: with the sign of (1.10) the coefficient of $\frac{(P \cdot r)(s \cdot r)}{r^4}$ is $d_\perp \xi \bar{\xi} - 2\bar{\xi} - 2\xi + 2 = d_\perp \xi \bar{\xi}$ and the UV coefficient comes out as $P_{gg}/2N_c$; with the extra minus it is $d_\perp \xi \bar{\xi} - 4$ and P_{gg} never appears. The sign of (1.10) is used below.

(ii) Θ_4 : the factor 2, again. Contracting the Θ_4 structure of (1.1), $-\frac{k^+ - p^+}{p^+} \left[\frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]$, with $\frac{P^i r^m}{P^2 r^2}$ gives $\frac{\bar{\xi}(s \cdot P) - (P \cdot r)}{P^2 r^2} \left[1 - \frac{r \cdot s}{2s^2}\right]$ — the $2s^2$ that is missing in (1.8) and in the expression you typed. Since you have now written it twice without the 2, everything below is carried with a parameter

$$V_4 = \frac{(s \cdot P) - (P \cdot r)}{P^2 r^2} \left[1 - \kappa \frac{r \cdot s}{s^2}\right], \quad \kappa = \frac{1}{2} \text{ (from (1.1))}, \quad \kappa = 1 \text{ (as printed)}.$$

Every result is given for both. Nothing that matters here depends on it: κ does not enter the UV coefficient, nor the rapidity block, nor the ξ integrals — only the finite remainder (§10D shows the UV coefficient is κ -independent to 7 digits).

(iii) The phase. Written out, $e^{-ik \cdot (w' - z)} e^{-ik \cdot (z - x)} = e^{-ik \cdot w'} e^{ik \cdot x}$ has no r -dependence at all, so (1.3) is consistent only if q is treated as ξ -independent, which is how (1.11) is written ($\int d\xi$ outside, $e^{iq \cdot r}$ inside). That is what is done below, and it is also what the paper needs: §4.2 has one power of $\ln(\Lambda/k^+)$ and hence $(\delta Y)^2$. If instead $q = \bar{\xi} k$ (the second exponential being $e^{-i\bar{\xi} k \cdot (z - x)}$), the ξ integrals below acquire a factor $e^{i\bar{\xi} k \cdot r}$: the ℓ_k terms are unchanged (they come from the endpoints, where the phase is $e^{ik \cdot r}$ and 1), but the finite ones stop being elementary and an extra $\ln(1/\bar{\xi})$ appears, generating ℓ_k^2 . See §9.

2. Variables, and the map to the paper

Keep your notation, $r = z - x$, $s = y - x$, $P = x' - w'$, $\xi = p^+/k^+$, $\bar{\xi} = 1 - \xi$. Then, as in (1.2),

$$p^+(y - x)^2 + (k^+ - p^+)(y - z)^2 = k^+ [\xi s^2 + \bar{\xi}(s - r)^2] = k^+ \bar{\xi} D, \quad D = (r - s)^2 + M^2, \quad M^2 = \frac{\xi}{\bar{\xi}} s^2.$$

Now put the paper's variables beside them, using $z_{\text{paper}} = x$, $z'_{\text{paper}} = z$:

paper (§2.8, §4.2)	here	paper	here
$Y = y - z$	$y - x = s$	Y^2	$s^2 \equiv B$
$Y' = y - z'$	$y - z = s - r$	$(Y')^2$	$(s - r)^2 \equiv A$
$Z = z - z'$	$x - z = -r$	Z^2	r^2
$Y^2 - (Y')^2$	$2s \cdot r - r^2 \equiv C$	$\ln \frac{Y^2}{(Y')^2}$	$L \equiv \ln \frac{s^2}{(s-r)^2}$

$$\bar{\xi} D = \bar{\xi} (Y')^2 + \xi Y^2 = \Delta(\xi) = A + \xi C, \quad C = B - A.$$

This is the denominator $((1 - \xi)(Y')^2 + \xi Y^2)$ of Eq. (4.16). Your $1/(\bar{\xi} D)$ and the paper's second energy denominator are the same object.

3. The bracket is the paper's vertex, contracted with one LO kernel

Strip the ξ -dependence off your six terms. Writing the bracket of your expression as

$$\text{bracket} = \xi \bar{\xi} V_1 + \frac{\xi}{\bar{\xi}} V_2 + \bar{\xi} V_3 + \frac{\bar{\xi}}{\xi} V_4 + \xi V_5 + V_6,$$

the six V_i are exactly your (1.4)–(1.10) with the prefactor removed:

$$\begin{aligned}
V_1 &= \frac{d_\perp}{2} \left[\frac{2(P \cdot r)(s \cdot r)}{P^2 r^4} - \frac{P \cdot r}{P^2 r^2} \right], & V_2 &= \frac{P \cdot s}{P^2 r^2} + \frac{(P \cdot s)(r \cdot s - r^2)}{2P^2 r^2 (s - r)^2}, \\
V_3 &= \frac{P \cdot r}{P^2 r^2} \left[2 - \frac{2(s \cdot r)}{r^2} - \frac{s \cdot r}{2s^2} \right], & V_4 &= \frac{(s \cdot P) - (P \cdot r)}{P^2 r^2} \left[1 - \kappa \frac{r \cdot s}{s^2} \right], \\
V_5 &= -\frac{P \cdot r}{P^2} \left[\frac{2(s \cdot r)}{r^4} + \frac{s^2 - s \cdot r}{2r^2 (s - r)^2} - \frac{1}{2r^2} \right], & V_6 &= \frac{2(P \cdot r)(r \cdot s)}{P^2 r^4} - \frac{P \cdot r}{P^2 r^2} + \frac{P \cdot s}{2P^2 s^2} \\
& & & - \frac{(P \cdot s)(r \cdot s)}{2P^2 r^2 s^2} + \frac{(r \cdot s)[(P \cdot s) - (P \cdot r)]}{2P^2 r^2 (s - r)^2}.
\end{aligned}$$

(That this reproduces the contraction of (1.1) exactly, with $\kappa = \frac{1}{2}$ and with the V_6 sign of (1.10), is checked symbolically in §10.)

Now compare with the paper. The vertex of (4.16), second factor, is

$$\underbrace{\frac{Y^2 - (Y')^2}{2Z^2}}_{\Theta^{jl}(y, z, z')} \delta^{jl} + \underbrace{\frac{1}{\xi} \left[\frac{(Y')^j Z^l}{Z^2} + \frac{Y^l (Y')^j}{2Y^2} \right]}_{\Lambda^{jl}(y, z, z')} + \underbrace{\frac{1}{1 - \xi} \left[\frac{Y^l Z^j}{Z^2} - \frac{Y^l (Y')^j}{2(Y')^2} \right]}_{\Lambda^{lj}(y, z', z) \text{ with } z \leftrightarrow z'}$$

in the notation (H.8). With $Y = s$, $Y' = s - r$, $Z = -r$ these are, structure for structure, your Θ_1 , your Θ_4 and your Θ_2 :

$$\begin{aligned}
\Lambda^{jl}(y, z, z') &\longleftrightarrow \frac{(s - r)^i (-r)^m}{r^2} + \frac{s^m (s - r)^i}{2s^2} && \text{(the bracket of your } \Theta_4), \\
\Lambda^{lj}(y, z', z) &\longleftrightarrow \frac{s^i (-r)^m}{r^2} - \frac{s^i (s - r)^m}{2(s - r)^2} && \text{(the bracket of your } \Theta_2).
\end{aligned}$$

The difference between your row and (4.16) is only *what the vertex is contracted with*. The paper contracts vertex $\times$ vertex, so the six products of $\{\Theta, \frac{1}{\xi}\Lambda, \frac{1}{1-\xi}\Lambda\}$ with themselves carry the weights $\xi\bar{\xi} \cdot \{1, \frac{1}{\xi}, \frac{1}{\xi}\} \otimes \{1, \frac{1}{\xi}, \frac{1}{\xi}\}$; your row contracts the vertex with the LO Weizsäcker–Williams kernel $\frac{(x' - w')^i (z - x)^m}{(x' - w')^2 (z - x)^2} = \frac{P^i r^m}{P^2 r^2}$, and the six weights collapse to

$$\left\{ \xi\bar{\xi}, \frac{\xi}{\bar{\xi}}, \bar{\xi}, \frac{\bar{\xi}}{\xi}, \xi, 1 \right\} = \xi\bar{\xi} \cdot \left\{ 1, \frac{1}{\xi^2}, \frac{1}{\xi}, \frac{1}{\xi^2}, \frac{1}{\xi}, \frac{1}{\xi\xi} \right\},$$

which is precisely the set that appears in (4.16) — this is the check that your row and the paper's are built from the same vertex. It is also why the p^+ integral needs nothing beyond (C.2), (C.5) and (C.6).

4. The p^+ integration: six elementary integrals

Everything ξ -dependent is now in the weights and in $\Delta(\xi) = A + \xi C$. Define, with $\lambda = \Lambda/k^+$, $\ell_k = \ln(1/\lambda) = \ln(k^+/\Lambda)$,

$$J_i \equiv \int_\lambda^{1-\lambda} d\xi \frac{w_i(\xi)}{\Delta(\xi)}, \quad w = \left(\xi\bar{\xi}, \frac{\xi}{\bar{\xi}}, \bar{\xi}, \frac{\bar{\xi}}{\xi}, \xi, 1 \right),$$

so that $\int d\xi \frac{\text{bracket}}{\xi D} = \sum_{i=1}^6 J_i V_i$. Derive them in the order in which they build on one another. Everything follows from

$$\frac{1}{\Delta} = \frac{1}{A + \xi C}, \quad \frac{1}{\xi \Delta} = \frac{1}{A} \left[\frac{1}{\xi} - \frac{C}{\Delta} \right], \quad \frac{1}{\bar{\xi} \Delta} = \frac{1}{B} \left[\frac{1}{\bar{\xi}} + \frac{C}{\Delta} \right], \quad \xi = \frac{\Delta - A}{C}.$$

J_6 (no weight). $\int_0^1 d\xi / \Delta = \frac{1}{C} \ln \frac{B}{A}$ — this is the paper's (C.2). Write $L \equiv \ln(B/A) = \ln[Y^2/(Y')^2]$:

$$J_6 = \frac{L}{C}.$$

J_5 ($w = \xi$). Using $\xi = (\Delta - A)/C$: $\int \frac{\xi}{\Delta} = \frac{1}{C} \int \left(1 - \frac{A}{\Delta} \right) = \frac{1}{C} - \frac{A}{C} J_6$, i.e.

$$J_5 = \frac{1}{C} - \frac{A}{C^2} L.$$

J_3 ($w = \bar{\xi}$). Simply $J_3 = J_6 - J_5$, and $1 + \frac{A}{C} = \frac{B}{C}$:

$$J_3 = \frac{B}{C^2} L - \frac{1}{C}.$$

J_1 ($w = \xi\bar{\xi}$). $J_1 = J_5 - \int \frac{\xi^2}{\Delta}$, and with $\xi^2 = \frac{(\Delta-A)^2}{C^2}$, $\int_0^1 \Delta d\xi = A + \frac{C}{2}$: $\int \frac{\xi^2}{\Delta} = \frac{1}{C^2} [\frac{C}{2} - A + \frac{A^2}{C} L]$, so

$$J_1 = \frac{1}{2C} + \frac{A}{C^2} - \frac{AB}{C^3} L.$$

J_4 ($w = \bar{\xi}/\xi$) — the first place a rapidity logarithm appears. From $\frac{\bar{\xi}}{\xi} = \frac{1}{\xi} - 1$ and the partial fraction above, $\int_{\lambda}^{1-\lambda} \frac{d\xi}{\xi\Delta} = \frac{1}{A} [\ell_k - L]$ — this is the paper's (C.5). Subtracting J_6 and using $\frac{1}{A} + \frac{1}{C} = \frac{B}{AC}$:

$$J_4 = \frac{\ell_k}{A} - \frac{B}{AC} L.$$

J_2 ($w = \xi/\bar{\xi}$) — the second rapidity logarithm. With $t = \bar{\xi}$, $\Delta = B - tC$, $\frac{1}{t(B-tC)} = \frac{1}{B} [\frac{1}{t} + \frac{C}{B-tC}]$, giving $\int_{\lambda}^{1-\lambda} \frac{d\xi}{\xi\Delta} = \frac{1}{B} [\ell_k + L]$. Subtracting J_6 and using $\frac{1}{B} - \frac{1}{C} = -\frac{A}{BC}$:

$$J_2 = \frac{\ell_k}{B} - \frac{A}{BC} L.$$

A sign in the paper. $\int \frac{d\xi}{\xi\Delta} = \frac{1}{B}(\ell_k + L)$ is the paper's (C.6), which is printed as $-\frac{1}{W^2}(\ln \frac{\Lambda}{k^+} + \ln \frac{W^2}{(W')^2}) = \frac{1}{B}(\ell_k - L)$. The L should come with the opposite sign: (C.6) is (C.5) with $\xi \rightarrow 1 - \xi$, which swaps $A \leftrightarrow B$ and hence $L \rightarrow -L$, turning $\frac{1}{A}(\ell_k - L)$ into $\frac{1}{B}(\ell_k + L)$. Direct quadrature confirms it (§10B): for $A = 1, B = 4$ the integral is 5.52739, not 4.83424. (C.5) itself is right.

5. The result of the p^+ integration

$$\begin{aligned} \mathcal{G}(r) &\equiv \int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi \frac{\text{bracket}}{\xi D} = \sum_{i=1}^6 J_i V_i, \\ J_1 &= \frac{1}{2C} + \frac{A}{C^2} - \frac{AB}{C^3} L, & J_2 &= \frac{\ell_k}{B} - \frac{A}{BC} L, & J_3 &= \frac{B}{C^2} L - \frac{1}{C}, \\ J_4 &= \frac{\ell_k}{A} - \frac{B}{AC} L, & J_5 &= \frac{1}{C} - \frac{A}{C^2} L, & J_6 &= \frac{L}{C}, \\ A &= (s - r)^2 = (Y')^2, & B &= s^2 = Y^2, & C &= B - A = 2s \cdot r - r^2, & L &= \ln \frac{B}{A}, & \ell_k &= \ln \frac{k^+}{\Lambda}. \end{aligned}$$

Regrouped by the object that multiplies each piece — which is how the paper presents it —

$$\begin{aligned} \mathcal{G} &= \ell_k \mathcal{G}_{\text{rap}} + L \mathcal{G}_{\text{log}} + \mathcal{G}_{\text{rat}}, \\ \mathcal{G}_{\text{rap}} &= \frac{V_2}{B} + \frac{V_4}{A} = \frac{V_2}{Y^2} + \frac{V_4}{(Y')^2}, \\ \mathcal{G}_{\text{log}} &= -\frac{AB}{C^3} V_1 - \frac{A}{BC} V_2 + \frac{B}{C^2} V_3 - \frac{B}{AC} V_4 - \frac{A}{C^2} V_5 + \frac{1}{C} V_6, \\ \mathcal{G}_{\text{rat}} &= \left(\frac{1}{2C} + \frac{A}{C^2} \right) V_1 - \frac{1}{C} V_3 + \frac{1}{C} V_5. \end{aligned}$$

5.1 What each block is, in the paper's language

$\ell_k \mathcal{G}_{\text{rap}}$ is Eq. (H.7). The paper's $(\delta Y)^2$ piece is

$$\Sigma_{JSSJ}^{(\delta Y)^2} \propto \left(\frac{\text{tr}(\Lambda(x, z, z') \Lambda^T(y, z, z'))}{(X')^2 (Y')^2} + \frac{\text{tr}(\Lambda(x, z', z) \Lambda^T(y, z', z))}{X^2 Y^2} \right) \ln \frac{\Lambda}{k^+},$$

i.e. the $1/\xi$ vertex divided by $(Y')^2$ plus the $1/(1-\xi)$ vertex divided by Y^2 . Ours is the same thing with one vertex instead of two:

$$\mathcal{G}_{\text{rap}} = \underbrace{\frac{V_4}{(Y')^2}}_{1/\xi \text{ vertex}} + \underbrace{\frac{V_2}{Y^2}}_{1/(1-\xi) \text{ vertex}} = \frac{(s \cdot P) - (P \cdot r)}{P^2 r^2 (s - r)^2} \left[1 - \kappa \frac{r \cdot s}{s^2} \right] + \frac{P \cdot s}{P^2 r^2 s^2} + \frac{(P \cdot s)(r \cdot s - r^2)}{2P^2 r^2 (s - r)^2 s^2}.$$

The correspondence is exact, term for term: it is the $1/\xi$ and $1/(1-\xi)$ poles of the vertex that produce ℓ_k , and each is divided by the value of $\Delta(\xi)$ at the endpoint where its pole sits — $\Delta(0) = A = (Y')^2$ for the $1/\xi$ pole, $\Delta(1) = B = Y^2$ for the $1/(1-\xi)$ pole.

$L \mathcal{G}_{\log} + \mathcal{G}_{\text{rat}}$ is Eq. (H.6). (H.6) has the same shape: a rational term $(-\frac{1}{2Z^4})$ plus terms carrying $\ln \frac{X^2}{(Y')^2}$ and $\ln \frac{Y^2}{(Y')^2}$. Ours has one such logarithm, $L = \ln \frac{Y^2}{(Y')^2}$, because there is one energy denominator.

Use $\sum_i J_i V_i$, not the regrouping, for numbers. $\mathcal{G}_{\log}$ and $\mathcal{G}_{\text{rat}}$ have spurious $1/C^3$ poles that cancel between them ($C \rightarrow 0$ is the circle $|r-s| = |s|$, where nothing happens physically). Each J_i separately is regular there, with

$$J_i = \frac{1}{B} \sum_{n \geq 0} \left(\frac{C}{B}\right)^n \int_{\lambda}^{1-\lambda} d\xi w_i(\xi) \bar{\xi}^n \implies \begin{array}{lll} J_1 B = \frac{1}{6} + \frac{\epsilon}{12} + \frac{\epsilon^2}{20} + \dots & J_2 B = (\ell_k - 1) + \frac{\epsilon}{2} + \dots & J_3 B = \frac{1}{2} + \frac{\epsilon}{3} + \dots \\ J_4 B = (\ell_k - 1) + \epsilon(\ell_k - \frac{3}{2}) + \dots & J_5 B = \frac{1}{2} + \frac{\epsilon}{6} + \dots & J_6 B = 1 + \frac{\epsilon}{2} + \dots \end{array}$$

with $\epsilon = C/B$. Evaluating J_1 from the closed form at $\epsilon \lesssim 10^{-5}$ loses all significance (§10G); use the series there.

6. The k^+ integration — δY and $(\delta Y)^2$

Your expression is differential in k^+ . To land on the form of §4.2, integrate over the rapidity slice as the paper does, $\int_{\Lambda}^{e^{\delta Y} \Lambda} \frac{dk^+}{k^+}$:

$$\int_{\Lambda}^{e^{\delta Y} \Lambda} \frac{dk^+}{k^+} = \delta Y, \quad \int_{\Lambda}^{e^{\delta Y} \Lambda} \frac{dk^+}{k^+} \ell_k = \int_{\Lambda}^{e^{\delta Y} \Lambda} \frac{dk^+}{k^+} \ln \frac{k^+}{\Lambda} = \frac{(\delta Y)^2}{2},$$

so that, exactly as in (4.18),

$$\int \frac{dk^+}{k^+} \mathcal{G} = \underbrace{\delta Y [L \mathcal{G}_{\log} + \mathcal{G}_{\text{rat}}]}_{\text{the NLO kernel, cf. (4.19), (H.6)}} + \underbrace{\frac{(\delta Y)^2}{2} \mathcal{G}_{\text{rap}}}_{\text{cf. (4.20), (H.7)}}.$$

The $(\delta Y)^2$ piece is the one that must reassemble into $\frac{1}{2}(\delta Y H_{\text{JIMWLK}}^{\text{LO}})^2$, Eq. (4.11) — and indeed $\mathcal{G}_{\text{rap}}$ is built only out of products of LO Weizsäcker–Williams kernels $\frac{r^i}{r^2}$, $\frac{s^i}{s^2}$, $\frac{(s-r)^i}{(s-r)^2}$, $\frac{P^i}{P^2}$, which is what (4.20) and (4.38) are made of.

7. UV, finite, and rapidity terms

Now separate. The only divergence left in $\mathcal{G}$ is at $r \rightarrow 0$, i.e. $z \rightarrow x$, where $\int d^2 z$ hits $1/r^2$. Every J_i is regular there ($C \rightarrow 0$), so the UV coefficient is $J_i(r=0)$ times the $r \rightarrow 0$ behaviour of V_i . In two dimensions $\langle \hat{r}^i \rangle = 0$, $\langle \hat{r}^i \hat{r}^j \rangle = \frac{1}{2} \delta^{ij}$, hence $\langle \frac{(P \cdot r)(s \cdot r)}{r^4} \rangle = \frac{s \cdot P}{d r^2}$ and $\langle \frac{P \cdot r}{r^2} \rangle = 0$:

	weight w_i	$V_i _{r \rightarrow 0}$ in units of $\frac{s \cdot P}{P^2 r^2}$	$B J_i(r=0) = \int d\xi w_i$	product
V_1 (Θ_1 , instantaneous)	$\xi \bar{\xi}$	$+\frac{d-1}{d} = +1$		$\frac{1}{6} \quad +\frac{1}{6}$
V_2 (Θ_2)	$\xi/\bar{\xi}$	$+1$	$\ell_k - 1$	$\ell_k - 1$
V_3 (Θ_3)	$\bar{\xi}$	-1	$\frac{1}{2}$	$-\frac{1}{2}$
V_4 (Θ_4)	$\bar{\xi}/\xi$	$+1$	$\ell_k - 1$	$\ell_k - 1$
V_5 (Θ_5)	ξ	-1	$\frac{1}{2}$	$-\frac{1}{2}$
V_6 (Θ_6)	1	$+1$	1	$+1$
total		$C_{\text{UV}}(\xi) = \xi \bar{\xi} + \frac{\xi}{\xi} + \frac{\bar{\xi}}{\bar{\xi}}$		$2\ell_k - \frac{11}{6}$

Two things to read off the table. Before the ξ integral, the $r \rightarrow 0$ coefficient is $\xi \bar{\xi} \cdot 1 + \frac{\xi}{\xi} \cdot 1 + \bar{\xi}(-1) + \frac{\bar{\xi}}{\bar{\xi}} \cdot 1 + \xi(-1) + 1 = \xi \bar{\xi} + \frac{\xi}{\xi} + \frac{\bar{\xi}}{\bar{\xi}} = \frac{P_{gg}(\xi)}{2N_c}$, because $-\bar{\xi} - \xi + 1 = 0$: $\Theta_3, \Theta_5, \Theta_6$ cancel among themselves in the UV, and $\xi(1-\xi)$ comes from Θ_1 , $\frac{\xi}{1-\xi}$ from Θ_2 , $\frac{1-\xi}{\xi}$ from Θ_4 . After the ξ integral the same cancellation leaves $2\ell_k - \frac{11}{6}$. κ never enters (§10D).

The separation. Exactly, for all r :

$$\mathcal{G}(r) = \underbrace{\left(2\ell_k - \frac{11}{6}\right) \frac{s \cdot P}{P^2 s^2} \frac{1}{r^2}}_{\text{UV } (z \rightarrow x)} + \underbrace{\ell_k \left[\mathcal{G}_{\text{rap}} - \frac{2}{s^2} \frac{s \cdot P}{P^2 r^2} \right]}_{\text{rapidity, finite at } z \rightarrow x} + \underbrace{\left[L \mathcal{G}_{\text{log}} + \mathcal{G}_{\text{rat}} + \frac{11}{6} \frac{s \cdot P}{P^2 s^2 r^2} \right]}_{\text{finite}},$$

the last two blocks being integrable at $r = 0$. The $2\ell_k$ of the UV coefficient comes entirely from the rapidity block and the $-\frac{11}{6}$ entirely from the other two; both are confirmed to six digits in §10E.

Only the first block needs a regulator. Doing its z integral, $\int d^2r e^{iq \cdot r} / (r^2 + m^2) = 2\pi K_0(m|q|) = \pi \log \frac{4e^{-2\gamma_E}}{m^2 q^2}$, gives

$$\left(2\ell_k - \frac{11}{6}\right) \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} \pi \log \frac{4e^{-2\gamma_E}}{m^2 q^2}, \quad -\frac{11}{6} \times 2N_c = -\frac{11N_c}{3} = -b|_{N_f=0},$$

a product of two Weizsäcker–Williams kernels times $b \log$ — the running-coupling structure of K_{JSJ} , Eq. (2.64)/(2.66).

8. The other order, and where $\frac{67}{18} - \frac{\pi^2}{3}$ comes from

If instead the z integral is done first (the companion note), the UV logarithm comes out carrying a $\bar{\xi}$ inside its argument, because $D_0 \equiv D|_{r=0} = s^2/\bar{\xi}$:

$$\frac{1}{\bar{\xi}} \frac{\mathcal{W}_{\text{UV}}}{P^2} = \pi \frac{P_{gg}(\xi)}{2N_c} \frac{(x' - w') \cdot (y - x)}{(x' - w')^2 (y - x)^2} \log \frac{(y - x)^2}{\bar{\xi} m^2},$$

and the ξ integral then needs the logarithmic moments of P_{gg} . These are exactly the paper's (C.10). Writing $C_{\text{UV}} = \xi \bar{\xi} + \bar{\xi} + \xi = \xi(1 - \xi) - 2 + \frac{1}{\xi(1 - \xi)}$ — which is *literally* the weight of (C.10) —

$$\int_{\lambda}^{1-\lambda} d\xi C_{\text{UV}} = 2\ell_k - \frac{11}{6}, \quad \int_{\lambda}^{1-\lambda} d\xi C_{\text{UV}} \ln [\xi(1 - \xi)] = -\ell_k^2 + \frac{67}{18} - \frac{\pi^2}{3},$$

the second being (C.10) with the $\ln(k^2/\mu^2)$ piece removed. Since C_{UV} is symmetric under $\xi \leftrightarrow \bar{\xi}$, $\int C_{\text{UV}} \ln(\xi \bar{\xi}) = 2 \int C_{\text{UV}} \ln \bar{\xi}$, so the moment this row actually needs is half of it:

$$\int_{\lambda}^{1-\lambda} d\xi C_{\text{UV}}(\xi) \log \frac{1}{1 - \xi} = \frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36} = -\frac{1}{2} \left(-\ell_k^2 + \frac{67}{18} - \frac{\pi^2}{3} \right),$$

giving, for the whole UV block of this row,

$$\int \frac{dp^+}{2\pi} \frac{P_{gg}(\xi)}{2N_c} \log \frac{k^+(y - x)^2}{(k^+ - p^+)m^2} = \frac{k^+}{2\pi} \left[\left(2\ell_k - \frac{11}{6}\right) \log \frac{(y - x)^2}{m^2} + \frac{\ell_k^2}{2} + \frac{\pi^2}{6} - \frac{67}{36} \right].$$

So: $-\frac{11}{6} \times 2N_c = -b$ at $N_f = 0$, and the constant is exactly the $\frac{67}{18} - \frac{\pi^2}{3}$ of (C.10) — the same combination that ends up as the $(\frac{67}{9} - \frac{\pi^2}{3})N_c$ of K_{JSJ} , Eqs. (2.64), (2.66), (4.36). (The remaining factor-of-two bookkeeping between $\frac{67}{18}$ and $\frac{67}{9}$ is the $x \leftrightarrow y$ symmetrisation of (4.31); the π^2 term additionally receives the self-energy/normalisation pieces of (4.31), which are not in this row, so this is an identification of the source of the constant, not a complete assembly of K_{JSJ} .)

Why the two orders look different. Doing ξ first (§5) the UV logarithm is $\log \frac{4e^{-2\gamma_E}}{m^2 q^2}$ with coefficient $2\ell_k - \frac{11}{6}$ and there is no ℓ_k^2 ; doing z first the logarithm is $\log \frac{(y-x)^2}{\xi m^2}$ and the extra $\log \frac{1}{\xi}$ generates $\frac{1}{2}\ell_k^2$. The totals agree; the $\bar{\xi}$ simply sits in a different block. The coefficient of $\log(1/m^2)$ — the only regulator-dependent number — is $2\ell_k - \frac{11}{6}$ in both.

9. What this reproduces, and what it cannot

Reproduced. The vertex and its six ξ -weights; the energy denominator $(1 - \xi)(Y')^2 + \xi Y^2$; the p^+ integration using nothing but (C.2), (C.5), (C.6); the resulting three-block structure (rapidity $\times \ln \frac{\Lambda}{k^+}$, a $\ln \frac{Y^2}{(Y')^2}$ block, a rational block) which is the shape of (H.6)–(H.7); the $\ell_k \rightarrow \frac{1}{2}(\delta Y)^2$ mechanism of (4.18)–(4.20); the fact that the $(\delta Y)^2$ block is built only from products of LO WW kernels, as (4.20) and (4.38) are; b and $\frac{67}{18} - \frac{\pi^2}{3}$.

Not reproducible from this row, and worth knowing before you spend more time on it. Σ_{JSJ} , Eq. (4.16), has two energy denominators, because both sides of the overlap are the two-gluon wave function $|\psi_{gg\rho}\rangle$. That is why H.2 needs

(C.3) and (C.4) as well, why both $\ln \frac{X'^2}{(X')^2}$ and $\ln \frac{Y'^2}{(Y')^2}$ appear, and where the $1/Z^4$ of K_{JSSJ} comes from. Your row has *one* energy denominator, one LO Weizsäcker–Williams kernel $\frac{P^i r^m}{P^2 r^2}$ in place of the second vertex, a Fourier phase, and a fifth point w' — it is a production object, $d^3 N/d^3 k$, not an S –matrix overlap. It therefore cannot give K_{JSSJ} directly.

The row it actually resembles is **§4.3, Σ_{JJSSJ} , Eq. (4.21)**: one energy denominator, the same vertex, contracted with products of LO kernels. Its ξ integration (4.22) is done with exactly (C.2), (C.5), (C.6) and yields exactly the structure obtained in §5 — a $\ln \frac{W^2}{(W')^2}$ block plus a $\ln \frac{\Lambda}{k^+}$ block whose coefficient is the $1/\xi$ vertex piece divided by $(W')^2$ plus the $1/(1 - \xi)$ piece divided by W^2 , which is our $\mathcal{G}_{\text{rap}} = V_4/(Y')^2 + V_2/Y^2$. If the target is a Σ_{JSSJ} –type kernel, the missing ingredient is the second energy denominator, i.e. the second $|\psi_{gg\rho}\rangle$ — not anything in the ξ or z integration.

10. Checks

All algebra symbolic (sympy) against the contraction of Eq. (1.1) itself; all integrals against quadrature.

```

kinematics: s = [ 0.7 -0.4]  P = [1.3 0.5]  s^2 = 0.6500  s.P = 0.7100  l_k = ln(k+/Lam) = 20.723266

A. the six xi-integrals J_i (analytic vs quadrature, lambda = 1e-9)
A=1.0 B=4.0
J1 analytic +0.072400835390 quad +0.072400835390 rel 7.7e-16
J2 analytic +5.065291929143 quad +5.065291949611 rel 4.0e-09
J3 analytic +0.282797493831 quad +0.282797492831 rel 3.5e-09
J4 analytic +18.874873355453 quad +18.874873359453 rel 2.1e-10
J5 analytic +0.179300626542 quad +0.179300626292 rel 1.4e-09
J6 analytic +0.462098120373 quad +0.462098119123 rel 2.7e-09
A=2.3 B=0.7
J1 analytic +0.118351965902 quad +0.118351965902 rel 5.9e-16
J2 analytic +27.161769629736 quad +27.161769751034 rel 4.5e-09
J3 analytic +0.299723106714 quad +0.299723106279 rel 1.5e-09
J4 analytic +8.783836003343 quad +8.783836003475 rel 1.5e-11
J5 analytic +0.443766935082 quad +0.443766933653 rel 3.2e-09
J6 analytic +0.743490041796 quad +0.743490039933 rel 2.5e-09
A=0.9 B=1.1
J1 analytic +0.167001436559 quad +0.167001436559 rel 9.0e-15
J2 analytic +18.018407006697 quad +18.018407081643 rel 4.2e-09
J3 analytic +0.518444125209 quad +0.518444124098 rel 2.1e-09
J4 analytic +21.799530013227 quad +21.799530014585 rel 6.2e-11
J5 analytic +0.484909352102 quad +0.484909351193 rel 1.9e-09
J6 analytic +1.003353477311 quad +1.003353475291 rel 2.0e-09

B. the paper's (C.5) and (C.6) [A=(W')^2=1, B=W^2=4]
(C.5) quadrature 19.3369714786
      paper (l_k - L)/A = 19.3369714758 agrees
(C.6) quadrature 5.5273900687
      paper (l_k - L)/B = 4.8342428690 DISAGREES (sign of L)
      correct (l_k + L)/B = 5.5273900495 agrees

C. the kernel G(r) = sum_i J_i V_i vs direct xi-quadrature of the whole bracket
r=[0.31 0.22] kappa=0.5: assembled +99.813352876 quad +99.813353193 rel 3.2e-09
r=[0.31 0.22] kappa=1.0: assembled +97.262612502 quad +97.262612819 rel 3.3e-09
r=[-0.5 0.8] kappa=0.5: assembled +14.920327198 quad +14.920327236 rel 2.6e-09
r=[-0.5 0.8] kappa=1.0: assembled +16.939098704 quad +16.939098743 rel 2.3e-09
r=[ 0.12 -0.05] kappa=0.5: assembled +1431.676306092 quad +1431.676309142 rel 2.1e-09
r=[ 0.12 -0.05] kappa=1.0: assembled +1371.521965492 quad +1371.521968538 rel 2.2e-09

D. UV coefficient of each vertex V_i (units of (s.P)/(P^2 r^2), angular average at r->0)
kappa=0.5 : [ 1. 1. -1. 1. -1. 1.]
kappa=1.0 : [ 1. 1. -1. 1. -1. 1.]
weighted sum = xi*xib + xi/xib + xib/xi = C_UV = P_gg/(2 Nc) (checked at xi=0.3, 0.65)

E. UV coefficient of the xi-integrated kernel, and how it splits over the blocks
kappa=0.5 : r^2<G> / [(s.P)/(P^2 s^2)] = +39.613189 predicted 2 l_k - 11/6 = +39.613198
      l_k block +2.000000 (=2) ln(B/A)+rational blocks -1.833335 (= -11/6)
kappa=1.0 : r^2<G> / [(s.P)/(P^2 s^2)] = +39.613182 predicted 2 l_k - 11/6 = +39.613198
      l_k block +1.999999 (=2) ln(B/A)+rational blocks -1.833334 (= -11/6)

F. the xi-integrals of C_UV (mpmath, 30 digits, lambda=1e-9) -- the paper's (C.10)
int C_UV dxi = 39.613198342559 2 l_k - 11/6 = 39.613198340559
int C_UV ln(xi xib) dxi = -429.02139290167 -l_k^2 + 67/18 - pi^2/3 = -429.02139286022
int C_UV ln(1/xib) dxi = 214.51069645084 l_k^2/2 + pi^2/6 - 67/36 = 214.51069643011
consistency I[ln(xi xib)] + 2 I[ln(1/xib)] = -7.6498848e-22

G. regularity of each J_i as C = B - A -> 0 (spurious 1/C, 1/C^2, 1/C^3 poles cancel)
C/B= 1.0e-02 : J1*B=0.16750503 (1/6) J3*B=0.50335854 (1/2) J5*B=0.50167505 (1/2) J6*B=1.00503359 (1)
      J2*B-l_k=-0.99498325 (-1) J4*B-l_k=-0.80585952 (-1)
C/B= 1.0e-04 : J1*B=0.16672983 (1/6) J3*B=0.50003333 (1/2) J5*B=0.50001667 (1/2) J6*B=1.00005000 (1)
      J2*B-l_k=-0.99995000 (-1) J4*B-l_k=-0.99807748 (-1)
C/B= 1.0e-06 : J1*B=65.98959961 (1/6) J3*B=0.49993451 (1/2) J5*B=0.50006599 (1/2) J6*B=1.00000050 (1)
      J2*B-l_k=-0.99999950 (-1) J4*B-l_k=-0.99998078 (-1)

```

Code

zint/src/xi_integration.py — symbolic check that the six V_i with weights $(\xi\bar{\xi}, \xi/\bar{\xi}, \bar{\xi}/\xi, \xi, 1)$ reproduce the contraction of (1.1) (true for $\kappa = \frac{1}{2}$, false for $\kappa = 1$). zint/src/xi_masters.py — the six J_i . zint/src/xi_assemble.py — $\mathcal{G}(r)$, the three blocks, the UV separation. zint/src/run_xi.py — every number above, written to zint/RESULTS_xi.txt.